

Literature review and experimental methods

In this chapter a review of the field of van der Waals' materials is presented. Graphene's properties and prospects are discussed and the motivation for van der Waals' heterostructures outlined. Characterisation of graphene and heterostructures by different methods will be reviewed, with particular focus on techniques around transmission electron microscopy (TEM). FIB sample preparation for TEM imaging is reviewed, with particular attention paid to the challenges of this method when applied to beam sensitive 2D materials and its limitations. Common image processing and analysis techniques as applied to cross section images of heterostructures are discussed in the penultimate section. The final section gives an overview of density functional theory (DFT) calculations for van der Waals' heterostructures. DFT was carried out by coauthors of papers in the following chapters but was not part of the project. Nevertheless, it is the only technique which provides useful comparison to the quantitative STEM data presented in subsequent chapters, and so an understanding of it's merits and shortcomings is useful.

2.1 Graphene, van der Waals' materials and van der Waals' heterostructures

It is perhaps a hallmark of success that the name “graphene” means different things to different communities: from a strictly free-standing single atomic plane of carbon, to a vast field of research encompassing all atomically thin crystals utilising van der Waals' bonding. For the purposes of this work, graphene is a two-dimensional (2D) allotrope of carbon: a single layer of

carbon atoms held together by sp^2 hybridised bonds to form a hexagonal lattice (see Section 2.1.1).[1] These sheets can be sequentially stacked together to form bilayer, trilayer e.t.c graphene up to bulk graphite[2], wherein each graphene sheet is known as a basal plane.[3] Van der Waals' materials can be defined as thick crystals composed of many layers of graphene or other 2D crystals, such as hexagonal boron nitride (hBN) and transition metal dichalcogenides (TMDCs). It should be noted that bulk graphite, and all bulk crystals with homogenous composition and van der Waals' bonding between atomically thin sheets, are included in the definition of van der Waals' materials. An interesting extension of this definition occurs when different 2D crystals are stacked upon one another to form van der Waals' heterostructures. We address these in the final part of this section. The papers presented in the following chapters characterise both van der Waals' materials and van der Waals' heterostructures, and not isolated sheets of 2D materials.

2.1.1 Graphene and van der Waals' heterostructures

In 1947 P R Wallace attempted to explain the electronic properties of graphite by considering the electronic structure of the individual graphitic basal planes, finding each “to be a semi-conductor with zero activation energy”.[4] This work captured the essence of the unusual properties a single graphitic sheet would have, though they proved to be far stranger.

Graphite is a three-dimensional (3D) allotrope of carbon, and the most stable form of solid carbon under standard conditions (other allotropes include diamond, fullerene and carbon nanotubes). It is the most well known van der Waals material due to its prevalent use as pencil lead, battery anodes

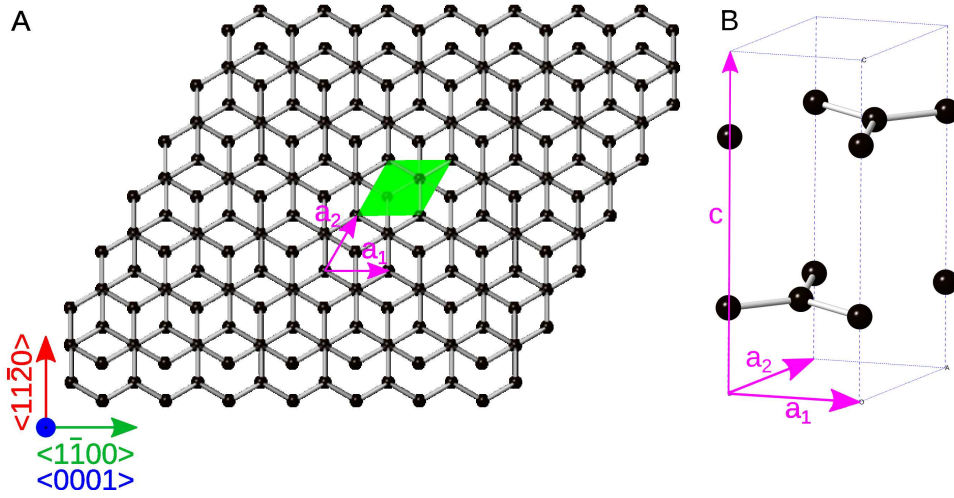


Figure 1: **A** Plan view schematic of graphite. The carbon atoms are shown as black spheres with grey covalent bonds connecting each atom to three neighbours to form the characteristic 'chicken wire' planar structure. Two planes can be seen stacked on top of one another, with the lower plane faded for clarity. Bottom left shows the highest symmetry crystal directions in graphite. The lattice vectors are drawn in purple and denoted $\mathbf{a}_1$ and $\mathbf{a}_2$, each corresponds to one lattice translation in the $\langle 1-100 \rangle$ or zigzag direction. The planar unit cell is highlighted green. **B** Oblique view of the graphite unit cell highlighted in **A**, which comprises of two planes of carbon atoms held together by weak van der Waals' bonding. The lattice vectors $\mathbf{a}_1$, $\mathbf{a}_2$ and $\mathbf{c}$ are drawn.

and solid state lubricant. Graphite has a hexagonal crystal structure with lattice constants of value $\mathbf{a} = 2.46 \text{ \AA}$ and $\mathbf{c} = 6.71 \text{ \AA}$ (see Figure 1B). The two planes of carbon atoms in the graphite unit cell are entirely covalently bonded in-plane and the two planes are held together by van der Waals' bonding. In crystallography these covalently bound single layers of carbon are labelled as "basal planes" when in graphite, defined as being perpendicular to the principle axis of the crystal (denoted as $\mathbf{c}$), and the primary plane down which the crystal can be cleaved. In the growing field of 2D materials, these planes are thought of as graphene layers stacked on top of one another. The most favourable stacking configuration in graphite is that shown in Figure 1, where half of the carbon atoms sit directly above the neighbouring sites and the other half sit in the centre of the hexagonal rings to give what is known as Bernal stacking.[3, 5] The distance between two basal planes is $\mathbf{c}/2 = 3.36 \text{ \AA}$. It is this large interlayer (basal plane - basal plane) distance, a feature definitive of these materials which results from the balance of forces comprising weak van der Waals' bonding[6], which allows atomically thin crystals to be cleaved and isolated from these materials.[7]

Graphene is an isolated basal plane of graphite, one atom thick and forming a hexagonal array of carbon atoms connected by strong covalent bonds (see Figure 2A,B). The valence electrons of each carbon atom are sp^2 hybridised, allowing an atom to bond to three nearest neighbour atoms and have electrons left over in an unhybridised $2p_z$ orbital for conduction across the planar lattice. (This can be contrasted with insulating diamond, where carbon is sp^3 hybridised and all valence electrons are engaged in bonding to the four nearest carbon atoms.) Finally, the two atoms in the unit cell of graphene belong

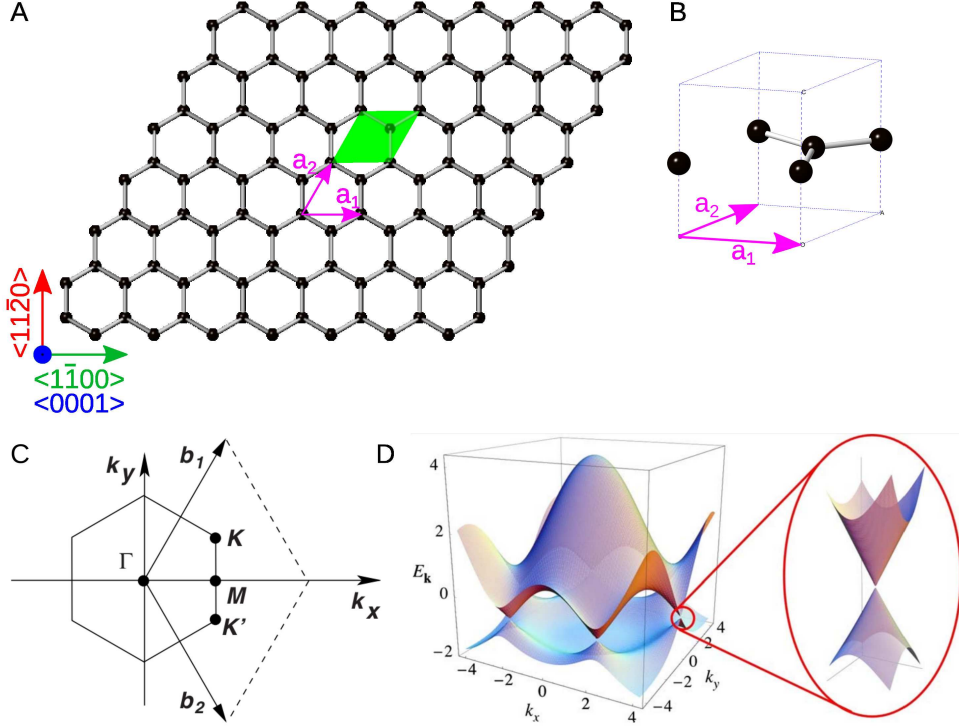


Figure 2: **A** Schematic showing an isolated sheet of graphene (a monolayer). The lattice unit vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are drawn as are the high symmetry directions in the crystal $\langle 11\bar{2}0 \rangle$, $\langle 1\bar{1}00 \rangle$ and $\langle 0001 \rangle$, which correspond to the armchair, zigzag and plane-normal directions, respectively. The unit cell of graphene is highlighted green and contains two carbon atoms in total. The highlighted unit cell is shown at an oblique angle in **B**, where the lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are also drawn. **C** is not drawn as isolated graphene is a 2D lattice and does not require a third lattice vector to describe it (unlike graphite). **C** The Brillouin zone of graphene with the reciprocal lattice vectors $\mathbf{b}_1$ and $\mathbf{b}_2$, and gamma, M, K and K' high symmetry points are labelled. **D** LEFT: The calculated dispersion of electronic states close to the Fermi level of graphene. RIGHT: The Dirac cones are located at the K and K' points with no curvature close to the Dirac point. Figure reproduced from Castro Neto et al.[8] Reprinted (with permission?) from American Physical Society, copyright 2009.

to two independent sublattices, so that any given atom is bonded only to those in the opposite sublattice. This has important implications for charge carriers in graphene. The band structure arising from the combination of these orbitals and the hexagonal sublattices is unique and underpins the exotic nature of charge carriers in graphene. Considering the reciprocal lattice of graphene (shown in Figure 2C) and using the tight binding approximation, the electronic dispersion of graphene can be plotted across the Brillouin zone. Where the conduction and valence bands meet at the K points, there are regions of linear dispersion which form a zero band gap about the Fermi level. Thus, graphene is a zero band gap semiconductor or a semimetal. These regions have been named “Dirac cones” and their effect on charge carriers moving through graphene, and resulting phenomena are explained next. One final note on the structure of graphene is that a fully relaxed structure holds static wrinkles which oscillate over tens of nanometre.[9] It is thought that this mechanism stabilises graphene (and other 2D materials), allowing it to exist as a single sheet of atoms.[10, 11]

In 2004 Nobel laureates Prof Andre Geim and Prof Konstantin Novoselov were the first to successfully isolate and characterise free standing graphene[1], although it should be acknowledged that monolayer graphite was known prior to this[12, 11]. They did this using the ‘Scotch tape’ method, where micromechanical cleavage is achieved using sticky tape to pull apart thick highly oriented pyrolytic graphite until it is atomically thin. The first electronic transport measurements not only showed that graphene’s band structure has Dirac cones at the K-points of the Brillouin zone, and that charge carriers close to the centre of the cone behave as massless Dirac fermions, but that

the anomalous quantum Hall effect and a non-trivial Berry's phase could be experimentally measured in high quality graphene flakes.[13, 14]

These phenomena all arise from two fundamental features of graphene's electronic structure. Firstly, charge carriers in graphene are chiral: Graphene has two sublattices which both contribute to the wavefunction of charge carriers, effectively burdening them with an additional quantum number "psuedospin". Psuedospin sets charge carriers in graphene apart from other two dimensional electron gases as this additional property suppresses the backscattering of carriers. Broadly, this means that exotic quantum mechanical phenomena are more likely to be observed in graphene than in more traditional condensed matter systems where scattering can dominate measurements. A surprising manifestation of this effect is the Klein paradox, where carriers will always propogate through a sharp barrier if the angle of incidence is 0° to the barrier normal.[15, 16, 17] Graphene also has a unique electronic bandstructure: a gapless, linear energy spectrum for electrons and holes.[2, 8] Charge carriers in normal semiconductors can be described by a relatively straightforward Schrödinger equation, where the energy and effective mass varies parabolically. In graphene the Schrödinger equation is not applicable to the linear energy spectrum. Instead the charge carriers must be described by a 2D relativistic Dirac equation as they are massless about the linear region; thus massless Dirac fermions. This is the unique case of monolayer graphene. In bilayer graphene an interesting combination of Dirac and Schrödinger equations results in massive Dirac fermions with four possible psuedospin states.

These revelations have propelled graphene to academic and mainstream fame, with the original Science paper recieving over 21,000 citations, placing it

in the top 100 most cited papers.[18] Rapid progress in the field revealed monolayer graphene to have carrier mobilities of up to $200,000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ at low temperature in the highest quality devices [19, 20], but, more interestingly for electronic devices, mobilities upwards of $15,000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ at room temperature[11]. It was not long before it was realised that the thin silicon dioxide substrate, which was relied upon to locate graphene under the optical microscope[21] and insulate it from the silicon back gate[13, 14], was breaking up the 2D electron gas in graphene through surface roughness, small areas of inhomogenous surface charging and impurities. The device performance was significantly improved by inserting a thin layer of insulating hexagonal boron nitride (hBN) between graphene and the substrate.[22] This heralded a step change in graphene device quality and was one of the first instances of a van der Waals' heterostructure.[23]

2.1.2 Hexagonal boron nitride: the white graphene

Monolayer hBN is an insulating isomorph of graphene, sometimes dubbed “white graphene”, with strong in-plane ionic bonds in place of graphene’s covalent ones.

Bulk hBN glues inert sheets of hexagonal boron and nitrogen lattices together with van der Waals’ bonds, similar to graphite. The lattice parameters of hBN have values of $\mathbf{a} = 2.50 \text{ \AA}$ and $\mathbf{c} = 6.66 \text{ \AA}$, which differ only slightly from graphite gives a lateral lattice mismatch of $\sim 1.6\%$. [24] hBN prefers AA’ stacking, where every boron (nitrogen) atom sits above a nitrogen (boron) atom of the neighbouring lattice (see Figure 3). [25] This stacking is highly unfavourable in graphite, which prefers AB (Bernal) stacking, yet the polar

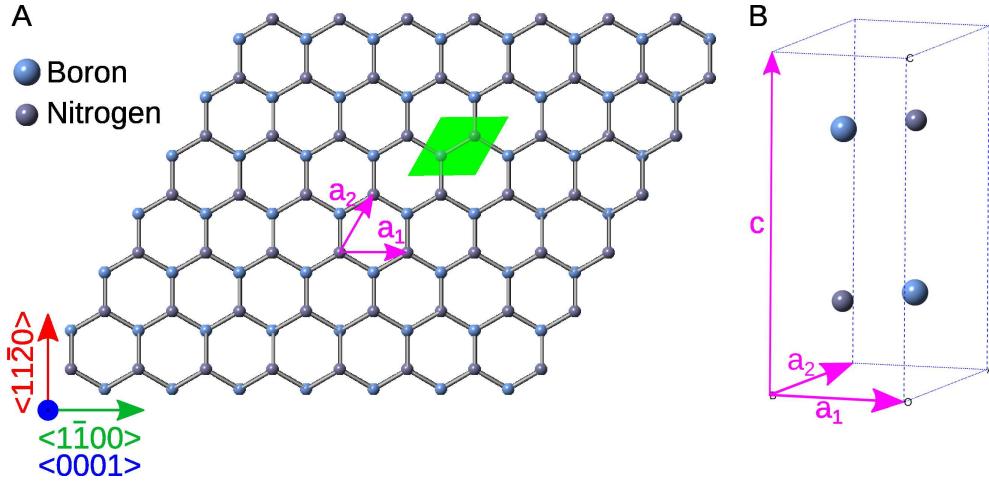


Figure 3: **A** Plan view schematic of hBN structure. Boron atoms are shown as larger blue spheres and Nitrogen atoms as smaller purple spheres. The 2D lattice is almost identical to graphene, differing only in elemental composition and lattice parameters. The lattice unit vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are drawn as are the high symmetry directions in the crystal $\langle 11\bar{2}0 \rangle$, $\langle 1\bar{1}00 \rangle$ and $\langle 0001 \rangle$, which correspond to the armchair, zigzag and plane-normal directions, respectively. The unit cell of hBN is highlighted green and contains one boron atom and one nitrogen atom. The highlighted unit cell is shown at an oblique angle in **B**, where the lattice vectors $\mathbf{a}_1$, $\mathbf{a}_2$ and $\mathbf{c}$ are also drawn.

nature of hBN bonding allows the $\mathbf{c}$ lattice parameters for both materials to be nearly identical.[26]

hBN is a dielectric and, unlike the majority of other 2D materials, this useful property does not require quantum confinement to be realised. This means that, as we will discuss later, hBN flakes of different thickness can be used to separate different conducting components of a heterostructure and prevent shorting.[27]. Furthermore, when thin sheets of graphene are placed on top of hBN and annealed, surface contaminants pool together in bubbles in what has become known as the self-cleaning phenomenon.[28, 29, 30] Self-cleaning arises from the van der Waals' attraction between hBN and graphene[31] and results in large areas of atomically clean interfaces, proving vital for high quality measurements.[22] These humble properties have allowed some of the most exciting physical phenomena to be realised, including negative differential conductance[32], Hofstadter's butterfly in bilayer graphene [33] and Klein tunneling of the chiral Dirac fermions.[15, 16, 17]

2.1.3 Other van der Waals' materials

There are an estimated 500 van der Waals' materials which hold the potential to yield large 2D flakes like graphene.[34] Their properties, and the effects of quantum confinement, are yet to be predicted for many of them. The 2D crystals that have been isolated since the discovery of graphene show a diverse spectrum of electronic properties including metals, semiconductors, insulators and superconductors. Some of the most recently isolated are silicene[35], germanene[36, 37], stanene[38], and phosphorene[39] (atomically thin layers of silicon, germanium, tin and phosphorus, respectively),

to name but a few. Furthermore, some of these 2D materials are expected to have various structural forms available when atomically thin. For example, phosphorene is predicted to have two different phases when monolayer; dubbed blue and black phosphorene, although only the latter has been realised experimentally.[40, 39, 41, 42] Many of these materials are unstable under standard conditions and so research has largely been limited to those that are stable (i.e. most of the transition metal dichalcogenides and some metal oxides)[23], although specialist handling strategies have been developed for unstable materials (see Section 2.1.5) [43].

One very promising class of 2D materials are the transition metal dichalcogenides (TMDCs). TMDCs are metallic or semiconducting crystals of formula MX_2 , where M is a transition metal (e.g. molybdenum, tungsten, niobium, etc.) and X is a chalcogen (sulfur, selenium or tellurium). Similarly to graphene and hBN, a TMDC has sheets of hexagonal crystal structure with strong in-plane bonding. Each sheet is stacked on top of another and held by van der Waals' forces. However, unlike graphene and hBN, they are not one atomic layer but three, with a sheet of transition metal atoms encapsulated by two sheets of chalcogen atoms. The most popular TMDCs are currently MoS_2 , WS_2 , $MoSe_2$ and WSe_2 due to them having an indirect semiconducting band gap in bulk crystals and a direct gap in monolayer crystals.[44, 45, 46] The direct band gaps of the popular semiconducting TMDCs range from roughly 1.5 - 2.1 eV[47], which correspond to near infrared - yellow photon energies in the visible spectrum. Furthermore, these monolayers are “multi-valley semiconductors”, meaning they have favourable and distinct localised states which a charge carrier wavefunction can occupy. These states appear

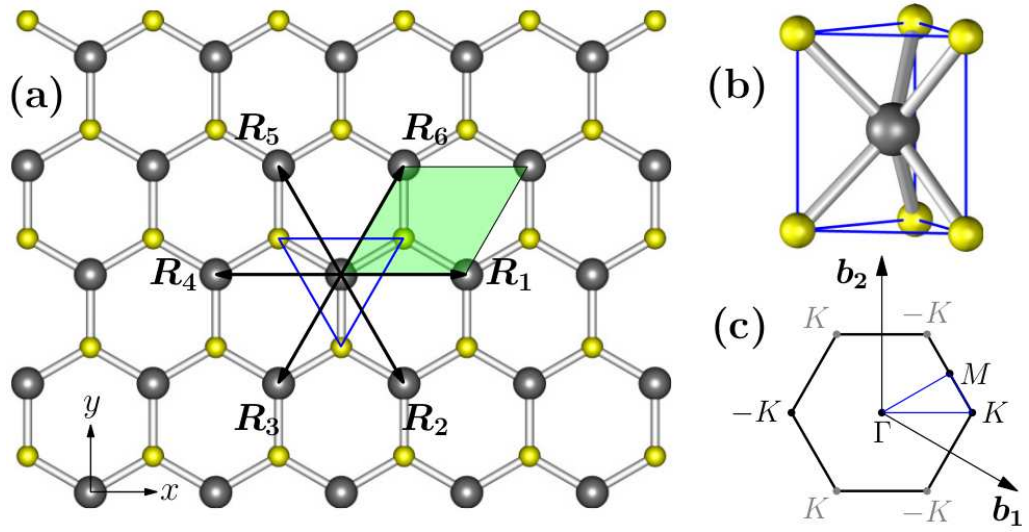


Figure 4: **a** Plan view schematic of a monolayer TMDC with chalcogen atoms shown as small yellow spheres and transition metal atoms shown as large grey spheres. The 2D unit cell is highlighted green and has lattice constant a . $\mathbf{R}_1$ to $\mathbf{R}_6$ shows the nearest neighbour transition metal atoms to the centre transition metal atom. **b** Oblique view of a metal atom with trigonal prismatic coordination to six nearest neighbour chalcogen atoms. The location of such a site is denoted by the blue triangle in A. **c** The Brillouin zone with basis vectors $\mathbf{b}_1$ and $\mathbf{b}_2$. The two valleys K and K^- are shown. [build your own figure..]

as valleys at the K points of the Brillouin zone (K and $-K$ in Figure 4), with differences between valleys arising from a lack of inversion symmetry in the crystal structure of monolayer TMDCs. A carrier can be confined to one of these valleys and so gains an extra constraint: a k -valley index (also known as valley degree of freedom or valley pseudospin). In addition to this, semiconducting TMDCs show strong spin-orbit coupling which arises from the valence electron states originating from d -orbitals. This results in spin up and spin down carriers having different energies as spin degeneracy is removed from the conduction and valence bands.

These phenomena manifest themselves in the form of optical selection rules

in semiconducting TMDC monolayers (although not present in bilayers or thicker). As with normal semiconductors, charge carriers can be promoted from the valence band to the conduction band. However, the spin states in each band are different for the K and -K valleys and can be selectively excited using left- and right-handed circularly polarised light.[48, 49] This selectivity arises from angular momentum conservation and time reversal symmetry constraints.[50]

Another attractive feature of semiconducting TMDCs is their exotic exciton physics. Promotion of carriers in monolayer semiconducting TMDCs cause electron hole pairs to form strongly bound exciton states due to the enhanced Coulomb interaction between the carriers that results from 2D confinement.[51, 52] These unique properties of semiconducting TMDCs allow a plethora of physical phenomena to be explored[53, 54, 55] and ultimately find application in ultrasensitive photodetectors[56, 57] and energy harvesting applications[58].

The second class of 2D TMDCs are metallic: niobium diselenide, niobium disulfide, tantalum diselenide and tantalum disulfide, although their band structure can potentially be made semiconducting if formed into nanoribbons.[59] Niobium diselenide is of particular interest currently as, if cooled sufficiently, charge density waves are observed.[60] These take the form of an ordered fluid of electrons that flow in a highly correlated but 'bumpy' manner through a material, and is thought to be linked to an unconventional form of superconductivity in layered materials.[61] Their occurrence in 2D materials has not been fully explained theoretically and so they are of great interest to fundamental research in condensed matter physics.[62, 63]

Bilayer (and thicker) TMDCs are capable of forming different stable structural polytypes (stacking orders) which often lead to modification of the electronic properties.[64] Furthermore, within these polytypes different polymorphs of the monolayers are possible i.e. where the metal atom is trigonally or octahedrally coordinated. Isolated monolayer TMDCs can have different polymorphs (metal atom coordination) but not polytypes as they are not stacked. In thicker TMDCs the most common polytypes are 1T, 2H and 3R, where each letter stands for trigonal, hexagonal and rhombohedral, respectively. The digits denote the number of monolayers in the unit cell. The dominant phase in a TMDC crystal is largely down to how the crystal was synthesised and its disposition towards a polymorph.[65, 64] For example, MoS₂ is most stable in the 2H phase with the metal atom trigonally coordinated, whereas TiS₂ takes the 1T phase where the metal is octohedrally coordinated. Multiple phases can even coexist within a flake and can arise during synthesis[66, 67] or patterned using electron beam irradiation[68]. Finally, azimuthal rotation of monolayers relative to one another creates variations in stacking via a moiré effect but does not seem to yield distinct and different polytypes [69].

2.1.4 Potential applications of graphene

In addition to graphene’s incredible electronic merits are a host of other properties which are unrivalled by other materials. It is the thinnest imaginable material (one atom thick)[1]; nearly completely optically transparent[70], has the largest specific surface area (2630 m² g⁻¹)[71]; the strongest and stiffest material ever measured (Young’s modulus of 1 TPa and intrinsic

strength of 130 GPa)[72]; has the highest thermal conductivity ($> 3,000 \text{ W mK}^{-1}$)[73]; the highest tolerance of electronic current density (a million times that of copper)[74, 75] and longest mean free path of electrons under ambient conditions[76]; the most stretchable crystal[77] and is impermeable to gases[78].

These myriad superior properties (most of which are only observed in the highest quality flakes) make graphene an ideal candidate for many applications.[79, 80] Graphene-based inks and paints are expected to provide conductive inks, antistatic and interference shielding, corrosion barriers and gas membranes.[81, 82, 83, 84, 85] Transparent conductive coatings for consumer flexible electronics (e.g. flexible touch screens and rollable electronic paper) are already being researched commercially.[86, 87, 88] By embedding graphene in polymer or ceramic matrices a new class of superstrong, lightweight and degradation resistant composites are anticipated to emerge over the next ten years.[89, 90, 91] Graphene based composites are also poised to play important roles in boosting the performance of many electrochemical and optoelectronic applications such as energy harvesting and storage technologies (fuel cells, supercapacitors, batteries, etc.).[92] The properties of graphene are extremely sensitive to environment and can be used in many different sensor applications. These range from drilling holes in graphene in order to sequence DNA[93, 94], to flakes acting as intrinsic strain gauges via Raman spectroscopy characterisation[95, 96]. Other sensing applications include gas environment, pressure and magnetic field, with devices possibly measuring many of these metrics simultaneously.[97, 98, 99, 100] Graphene is easily functionalised[101, 102] yet relatively inert which opens up possibilities in

bioapplications, particularly in biomolecular sensing, tissue engineering, regenerative medicine and drug delivery.[103, 104, 105]

It is widely anticipated that graphene will make its biggest impact in the fields of electronics and photonics.[8, 106, 107, 108, 109] The poignant absence of a band gap in monolayer and bilayer graphene all but eliminates its application as a speedy and direct replacement for silicon in conventional logic gate architectures. Nevertheless, there are currently three strategies to open a band gap in graphene: confining the width of graphene to form nanoribbons[110, 111], biasing bilayer graphene[112, 113, 114], and straining graphene[115, 116]. Monolayer graphene nanoribbons currently show the most promise with bandgaps scaling inversely with the width of the ribbon, in some cases reaching 200 meV for < 20 nm widths[110, 111]. However, amongst problems relating to reproducible ribbon quality, confined carriers rapidly lose mobility and risk negating any advantages over silicon. The main motivation for integrating graphene in logic devices is to extend the trend in integrated circuit advancement as professed by Moore's Law, which in turn forms the basis of financial growth for the semiconductor industry.[117, 118] Through reduction in the size of silicon metal-oxide-semiconductor field-effect transistors (MOSFETs) the performance of integrated circuits has roughly doubled every 18 months for the last fifty years.[107] However size reduction is rapidly approaching its limit in silicon and new materials are being sought to increase device performance and decrease feature size.[119] Although mobility is typically cited as graphene's most attractive feature for high speed logic devices, it is actually the potential to have robust single atom channel widths which sustains interest within the semiconductor community (despite

a lack of bandgap).[107] New device concepts are being explored to work around graphene's shortcomings, the most promising of which is the bilayer pseudospin FET; two graphene monolayers separated by a similarly thin dielectric. To switch the device on a gate bias is applied which encourages carriers to tunnel through the dielectric, effectively creating a short circuit current to flow through the channel. Under different bias conditions the tunnelling resistance is high, suppressing tunnelling and switching the device off.[120] This is anticipated to yield high on-off ratios with a very low switching energy.

The unique optical properties of graphene make it an excellent candidate material for many photonic devices. Photodetectors, mode-locked lasers, polarizers, isolator devices and optical modulators are all anticipated applications for graphene's optical properties, many of which already have proof of concept devices.[121, 122, 123, 97, 124, 125, 126, 127, 128, 129, 130] Optical modulators are key components in photonics systems as they are used to control the characteristics of light packets carrying signals between devices.[131] Modulation speed, the spectral range of the modulator, energy consumption and device size all dictate the quality of an optical modulator, and all of which depend on modulating material. Graphene is particularly useful for optical modulation for three reasons: Graphene has a high modulation speed owing to its intrinsically high carrier mobility; the absorption and modulation characteristics can be directly controlled by changing the Fermi level using a gate bias; and graphene has a large optical bandwidth with a constant absorption of 2.293% from the infrared to visible wavelengths (which happens to coincide with the optical communication bandwidth required).[132, 126] This absorp-

tion may seem low, however when taken into consideration with graphene’s single atom thickness it is in fact very high compared to other photonics materials. These properties could prove most lucrative in terahertz modulators and detectors, which would begin to bridge what has become known as the “terahertz gap” and pave the way to cheap and powerful long wavelength applications.[122, 133, 129]

The main bottlenecks to progress in all these areas lie in the fundamental problems of handling graphene and scaleable synthesis.

2.1.5 Making and transferring graphene flakes

The papers presented in the following sections of this thesis use devices made from mechanically exfoliated 2D flakes. For context a brief review of other synthesis routes to graphene flakes are presented here.

The myriad different methods for synthesising graphene flakes fall broadly under mechanical exfoliation, chemical growth and liquid phase exfoliation. Exfoliation methods are considered “top-down” synthesis routes as they directly remove graphene sheets from a bulk crystal of graphite. Chemical growth methods are “bottom-up” as they typically use a carbon-based precursor gas and favourable growth conditions (a catalyst, heat, pressure-control etc) to form sheets of graphene on a surface. The relative merits of each of these methods are best judged in the context of their target application. For example, liquid exfoliation methods are best suited to inexpensive large-volume production of graphene suspensions to generate conductive surfaces, composites and barrier films.[134] However, liquid exfoliation struggles to make atomically thin, clean, highly crystalline, large area flakes required for elec-

tronic and photonic devices (this is not for lack of trying[135]). Nevertheless, liquid exfoliation methods are easily the cheapest and most scalable syntheses for graphene (and other 2D materials), and as such the applications which use liquid exfoliated graphene will be the first to be commercialised.[79, 80] Liquid exfoliation methods generally disperse bulk flakes in solvent and use ultrasound to induce bubble cavitation, which applies stresses to the bulk flakes causing them to cleave and fragment.[136] Different centrifuging methods have shown to be size and shape selective[137] and other liquid exfoliation strategies also use ion intercalation to cleave sheets of graphene.[138, 139]

The highest quality devices, used for prototyping, proof-of-concept and exploring new physical phenomena, are still made from mechanically exfoliated flakes. Mechanical exfoliation is highly inefficient in terms of materials and time. Only a few suitable flakes are produced from a bulk crystal of graphite and, whilst the exfoliation step is not very time consuming, manually searching for flakes in an optical microscope is.[136] Furthermore, the size, shape and crystal directions of graphene flakes are not controllable by this method.[7] Chemical growth methods are already showing great promise for the controlled growth of graphene and other 2D materials on different surfaces. Chemical growth can be roughly split into two principle methodologies: silicon-carbide (SiC) epitaxy and chemical vapour deposition (CVD). In the former, a crystalline wafer of SiC is heated in excess of 2000°C in an inert atmosphere. The silicon atoms at the lowest energy surface sublime and the remaining carbon rearranges to graphene. This method offers fine control of graphene thickness on an insulating substrate and has been shown to synthesise high quality flakes.[140, 141] CVD is a very versatile technique

as not only does it offer controlled deposition of graphene there are also a large variety of surfaces on which it can be grown[7], the best of which is currently copper for monolayer graphene[142, 143]. Graphene is either grown from a carbon-based gas precursor on a catalytic metal surface, or by surface segregation of carbon precipitating from solution in the metal bulk (both mechanisms can coexist, e.g. graphene synthesised on nickel in methane[144]). Despite the bottom-up nature of this method there are a few disadvantages to overcome before this supplants mechanical exfoliation for device prototyping. The first, and biggest problem, is that the electronic properties of a flake cannot be tested whilst on a metallic substrate, requiring an additional transfer step.[145, 146, 147] The second disadvantage is that multiple nucleation sites on the catalyst surface lead to many graphene crystallites growing simultaneously. At the end of the reaction, when the catalyst surface is completely covered, these crystallites interface with grain boundaries between azimuthally rotated graphene grains.[7] Grain boundaries are thought to disrupt the ballistic transport of carriers through graphene and degrade the quality of the electronic properties.[148, 149] More recently, large single crystals (> 0.4 mm across) have been grown under very carefully controlled conditions.[150, 151]

Almost regardless of how graphene is made, poor handling rapidly degrades flake quality as crystal defects, wrinkles, tears and impurities set in. Indeed, careful handling of CVD grown graphene has electronic properties to rival that of mechanically exfoliated graphene.[152] Handling is most likely the next key hurdle to high quality graphene being realised commercially. Strategies rely on the differing cohesion between materials, with transfer techniques to and

from different surfaces developed to aid the rapid but controlled transfer of graphene sheets from synthesis to target surfaces. The majority of these techniques are developed for SiC epitaxially or CVD grown graphene.[153, 146, 147] Nevertheless, many of the transfer techniques developed for CVD grown 2D materials are possible candidate transfer techniques for fabricating van der Waals' heterostructures.[152]

In the following sections the two most popular transfer methods (at time of writing) are described: wet and dry transfer techniques. These are used in to fabricate the heterostructures presented in the following chapters.

2.1.6 van der Waals heterostructure fabrication

There are two popular methods used to transfer one mechanically exfoliated 2D crystal on top of another: dry transfer and wet transfer, although other techniques are possible for different synthesis routes and may replace these soon.[147] Wet transfer uses a sacrificial water soluble polymer layer to transfer 2D flakes. The sacrificial layer is spin coated onto a silicon substrate and then further coated with polymethyl-meth-acrylate (PMMA). Graphite is mechanically exfoliated onto this polymer stack and thin flakes identified by optical microscopy[21] and Raman spectroscopy[154]. To isolate a suitable flake, the stack is floated on the surface of a water bath until the water soluble polymer dissolves, allowing the silicon substrate to sink. A glass slide with a cut hole in the middle is then maneuvered with a micromanipulator over the flake on the floating PMMA to remove them from the bath, all observed under an optical microscope. The PMMA is flipped over so that the flake-side is facing the bench. The flake is aligned above a target substrate flake

(e.g. hBN) and the two brought into contact to make a “stack”. The PMMA is then dissolved with acetone and washed away. The stack is annealed to $> 110\text{ }^{\circ}\text{C}$ to drive off solvents and contamination.

Dry transfer is a variation on the wet transfer technique which attempts to minimise contamination between flakes and greatly speed up the transfer process.[155, 156, 157] The wet transfer procedure is followed to the point where a flake is isolated on a PMMA membrane above a smaller target flake. Once the two flakes are in contact, instead of dissolving the PMMA, the PMMA is lifted from the substrate bringing the two flakes with it. This step is then repeated for many flakes to build up a bespoke complex stack of different 2D materials. On the last transfer step, the PMMA is dissolved with acetone and the stack annealed. Through this method the majority of interfaces between flakes will be isolated from PMMA and solvents, greatly reducing the PMMA residue present.[153, 156, 152]

Stacks of 2D crystals are converted into devices by adding metallic contacts to the surface[32, 158, 159] or edges of graphene sheets[156, 157]. More recently, in the latest generation of devices, metal contacts are placed on the surface of an encapsulating hBN layer so that electrons can be injected by tunnelling into buried conductive layers or used as a gate to probe the electronic properties of a device.[27, 160, 161, 30, 135, 43]

2.1.7 van der Waals’ heterostructures

Band structure (or band gap) engineering of semiconductors underpins modern electronics[162, 163], and has lead to breakthrough applications such as high mobility MOSFET transistors and light emitting diodes (LEDs)[164].

Band structure engineering is traditionally achieved by growing different semiconductor crystals sequentially to form an extended layered superlattice. When a charge carrier travels from one crystal to another, it is influenced by the host crystal's band structure. This effect can be used to create quantum wells, where electrons and holes become confined within a small region of the superlattice where the band gap is smallest. Confinement of carriers allows exotic physical phenomena to be explored, for example exciton states and photoluminescence.[165] The fabrication of these superlattices is not trivial. For example, growth methods for traditional heterostructures use expensive bottom-up techniques such as molecular beam epitaxy (MBE) and pulsed laser deposition which require high vacuum conditions[166]. Furthermore, the choice of materials is severely limited; the semiconductor crystals need to have very similar lattice constants as the layers are strongly bound to one another, however the electronic properties can be fine tuned with doping.[167, 168]

In contrast, van der Waals' heterostructures fabricated from mechanical exfoliated flakes are comparatively straightforward. The different 2D materials do not need to have matching lattice constants due to their top-down assembly[23]; the 2D flakes can be randomly orientated to one another and still produce interesting band structures[157, 155], or they can be aligned with high angular precision to explore exotic physical phenomena[33, 169, 24]; the handling of the majority of 2D materials does not require epitaxy methods within a complex high vacuum system[170, 7, 171, 172, 173]; and those 2D materials which are sensitive to air can be encapsulated within other 2D materials to produce inert heterostructures with novel properties[43, 174, 175].

The unprecedented control over the handling and characterisation of 2D materials within van der Waals' heterostructures has lead to them being used in conjunction with other materials such as nanomaterials[176, 177, 135], traditional semiconductor materials[178, 179, 180, 175], self-assembled monolayers[181, 182, 183], and even biological molecules[184], many of which use complex device geometries depending on their intended application.[185]

The majority of van der Waals' heterostructures use vertical transport of charge carriers between layers. However there are also lateral heterostructures which use planar transport across grain boundaries between different 2D materials.[186, 187, 188] They are comparatively complex to synthesise, requiring carefully controlled growth conditions and have similar constraints to traditional semiconductor heterostructures.[189] They are not discussed further in this thesis.

Vertical van der Waals' heterostructures mostly benefit from a particularly humble mechanism: carrier tunneling through hBN. Carrier injection into early van der Waals' heterostructure devices was usually carried out by adding gold contacts to the surface or edges of graphene.[156, 157] This technique works reasonably well, however the contact resistance and damage to the graphene is difficult to control. It was later found that carriers could be injected into the heterostructure through an insulating layer of hBN.[190] This came with some advantages as the thickness of the hBN determined the tunneling resistance of the contact[27], the buried conductive sheets were protected from the gold contact by the hBN, and the surfaces self-cleaned[155]. Furthermore, quantum tunneling through hBN preserves the spin state of the carriers allowing spin transport in graphene to be explored using ferromag-

netic contacts.[191, 192] hBN also allows different components to be isolated from one another until high enough potential enduces quantum tunneling of carriers between components.[190, 27] This has lead to some of the highest on/off ratios in 2D material devices[179, 30, 193, 160] as well as allowing the electronic properties of unstable materials to be probed[43, 174].

Van der Waals' heterostructures are not limited to layering different 2D materials; stacking materials of varying dimensional confinement is also of great interest to electronics and nanofluidics thanks to the fine control and cleanliness of these structures.[194, 195, 185, 196] Fundamental experimental work on water flow phenomena in nanoscale capillaries using a graphite-based heterostructure is presented in Chapter 5.

Finally, a surprising application of van der Waals' heterostructures is in TEM imaging of 2D materials.[197, 198] By encapsulating a 2D material in graphene the enclosed material can be studied with higher doses of electrons without structural damage. Higher doses of electrons increase the signal to noise ratio when acquiring an image or spectroscopic data.[199] Previous efforts to image 2D materials were limited by structural damage and rapid formation of defects under the electron beam.[200, 201, 202, 203] Graphene provides rapid charge transfer away from the irradiated site, limiting radiolysis effects, and also protects the 2D material from mobile contaminants that can act as etchants.[197, 198] This is a highly successful method for imaging 2D materials, however filtering of the fourier transform of both graphene lattices is required to isolate the structure of interest. This results in the potential loss of useful information about the target lattice and often image simulation is required to interpret the data.[204] Nevertheless, this method has been ap-

plied to many different 2D materials to study their intrinsic defects[205][Lan’s paper?] and most recently *in situ* microscopy of liquid systems[196, 206].

2.1.8 Characterisation of van der Waals’ heterostructures

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The primary method for characterising a van der Waals’ heterostructure the electronic transport measurements or the measurement of a physical phenomenon. For the most part this is due to van der Waals’ heterostructures being relatively new and so standards in quality have not been established. Characterisation by this method relies on the sensitivity of the phenomenon being measured, for example simple electron mobility measurements are less sensitive to structural defects and impurities than charge density wave propagation. For now any evidence of a device working satisfies the burden of proof for its purpose (i.e. publication). This is beginning to change as the requirement for device reproducibility and standards comes into demand.

Common characterisation techniques for probing device structure and quality are atomic force microscopy (AFM), scanning near-field optical microscopy (SNOM), and Raman spectroscopy. These are often used not only on heterostructures but on the isolated 2D material flakes prior to assembly.

2.2 (S)TEM imaging of 2D materials

2.2.1 TEM imaging

2.2.2 STEM imaging

2.2.3 Analytical spectroscopic methods: EDXS and EELS

2.2.4 Advanced signal processing techniques for spectrum image datasets

2.2.5 Electron beam induced effects: Damage, heating and contamination

2.2.6 FIB Section (try to remain objective)

2.2.7 Cross sectional (S)TEM imaging of 2D materials

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